

1. What is wrong with this Python loop:

1 / 1 point

```
1 n = 5
2 while n > 0 :
3     print(n)
4     print('All done')
```

Select the correct option.

- ☒ This loop will run forever.
- ☐ The `print('All done')` statement should be indented four spaces.
- ☐ There should be no colon on the `while` statement.
- ☐ `while` is not a Python reserved word.

✔ Nice work

Correct! The loop will run forever because the value of `n` is never decremented within the loop.

2. What does the `break` statement do?

1 / 1 point

Select the correct option.

- ☐ Exits the program
- ☒ Exits the currently executing loop
- ☐ Jumps to the "top" of the loop and starts the next iteration
- ☐ Resets the iteration variable to its initial value

✔ Nice work

Correct! The `break` statement immediately exits the loop in which it is placed.

3. What does the `continue` statement do?

1 / 1 point

Select the correct option.

- ☐ Resets the iteration variable to its initial value
- ☐ Exits the program
- ☒ Jumps to the "top" of the loop and starts the next iteration
- ☐ Exits the currently executing loop

✔ Nice work

Correct! The `continue` statement skips the rest of the loop iteration and starts the next iteration at the top of the loop.

4. What does the following Python program print out?

1 / 1 point

```
1 tot = 0
2 for i in [5, 4, 3, 2, 1] :
3     tot = tot + 1
4     print(tot)
```

Select the correct option.

- ☐ 10
- ☐ 0
- ☒ 5
- ☐ 15

✔ Nice work

Correct! The loop increments `tot` for each item in the list, resulting in a final value of 5.

5. What is the *iteration* variable in the following Python code:

1 / 1 point

```
1 friends = ['Joseph', 'Glenn', 'Sally']
2 for friend in friends :
3     print('Happy New Year:', friend)
4     print('Done!')
```

Select the correct option.

- ☒ friend
- ☐ Glenn
- ☐ Sally
- ☐ friends

✔ Nice work

Correct! `friend` is the iteration variable that takes each value in the list `friends`.

6. What is a good description of the following bit of Python code?

1 / 1 point

```
1 zork = 0
2 for thing in [9, 41, 12, 3, 74, 15] :
3     zork = zork + thing
4     print('After', zork)
```

Select the correct option.

- ☒ Sum all the elements of a list.
- ☐ Find the largest item in a list.
- ☐ Count all of the elements in a list.
- ☐ Find the smallest item in a list.

✔ Nice work

Correct! The code sums all the elements in the list, storing the result in `zork`.

7. What will the following code print out?

1 / 1 point

```
1 smallest_so_far = -1
2 for the_num in [9, 41, 12, 3, 74, 15] :
3     if the_num < smallest_so_far :
4         smallest_so_far = the_num
5     print(smallest_so_far)
```

Hint: This is a trick question and most would say this code has a bug. So, read carefully!

Select the correct option.

- ☒ -1
- ☐ 74
- ☐ 42
- ☐ 3

✔ Nice work

Correct! If `smallest_so_far` starts at -1, the if condition will never be true, as no list elements are smaller than -1.

8. What is a good statement to describe the `is` operator as used in the following if statement:

1 / 1 point

```
1 if smallest is None :
2     smallest = value
```

Select the correct option.

- ☐ Is true if the `smallest` variable has a value of -1
- ☒ matches both type and value
- ☐ Looks up `'None'` in the `smallest` variable if it is a string
- ☐ The `if` statement is a syntax error

✔ Nice work

Correct! The `is` operator checks for both type and value equality without type conversion, making it stricter than `==`.

9. Which reserved word indicates the start of an "indefinite" loop in Python?

1 / 1 point

Select the correct option.

- ☐ def
- ☐ indef
- ☐ for
- ☒ while

✔ Nice work

Correct! The `while` keyword is used to create loops that continue until a condition becomes false.

10. How many times will the body of the following loop be executed?

1 / 1 point

```
1 n = 0
2 while n > 0 :
3     print('Lather')
4     print('Rinse')
5     print('Dry off!')
```

Select the correct option.

- ☒ 0
- ☐ This is an infinite loop
- ☐ 1
- ☐ 5

✔ Nice work

Correct! The loop body will never execute because the condition `n > 0` is false from the start.